

ECU-25 — Signal Map

Every quantity the controller measures, every output it drives, and exactly what it commands on the engine ECU.

Target set: Kirloskar KG4-25WS1 — 25 kVA, 415 V, 50 Hz, engine 3R550ETA 4G1 (3-cylinder 1.65 L CRDi, CPCB IV+, EGR + DOC), 12 V / 75 Ah electrical system.

Derived from the schematic generator (`hardware/kicad/gen/gen_ecu25.py`), the firmware types (`firmware/core/gcu_types.h`), and the two field protocols. Component values trace to the reviewed calculation sheet, `docs/circuits/README.md`.

Group	Count	Detail
Analog measurement channels	14	6 AC voltage, 3 AC current, 3 engine senders, battery, D+
Frequency input	1	magnetic pickup
Digital inputs	8	5 assigned, 3 spare
Engine data over J1939	5 PGNs	rpm, oil, coolant, fuel, DM1 lamps + DTC
Operator keys	3	start, stop, mode
Relay outputs	6	all dry contacts
Other outputs	8	D+ excitation, backlight, SPI selects, RS485 DE, LED
Alarms evaluated	21	every 10 ms tick

Two things this map makes explicit, because they are the most common misunderstanding about this class of controller:

- Most engine data has two possible sources.** On this engine, rpm / oil pressure / coolant temperature / fuel level arrive over **J1939** from the engine's own ECU. The analog sender front end is the **legacy path**, kept for mechanically-governed engines and as a cross-check. The control logic never knows which source filled the value: the runtime **merges** them, so a fresh J1939 value wins and a stale one leaves the sender reading standing. An unconditional overwrite would wipe four good readings the moment the bus went quiet — and permanently on a legacy engine with no ECU.
- ECU-25 does not run the engine. It supervises it.** Over CAN it is a listener only: it transmits nothing but its own address claim. Its entire physical authority over the engine is one relay contact — run-enable — plus a starter pilot. See §8.

1. AC electrical measurement

Six voltage channels and three current channels, all sampled together against the buffered VREF_MID 1.65 V bias, so the ADC sees a signed waveform on a single supply.

#	Quantity	Sensor / source	Terminal	Measuring range	Front end	Scaling	MCU pin
1	Generator L1-N volts	direct connection to alternator terminals	J5-1	0-276 V RMS (clips), nominal 240	4 × 330k + 5.62k divider, biased to 1.65 V, 1.1 kHz filter	÷235.9	PC0
2	Generator L2-N volts	"	J5-2	"	"	÷235.9	PC1
3	Generator L3-N volts	"	J5-3	"	"	÷235.9	PC2

#	Quantity	Sensor / source	Terminal	Measuring range	Front end	Scaling	MCU pin
4	Mains L1-N volts	direct connection to the utility side	J6-1	"	"	÷235.9	PC3
5	Mains L2-N volts	"	J6-2	"	"	÷235.9	PC4
6	Mains L3-N volts	"	J6-3	"	"	÷235.9	PC5
7	Load current L1	current transformer on the phase cable	J7-1 / J7-2	0-117 A primary with 50:5	0.05 Ω burden → MCP6002 × 2, phase-matched filter	1 A sec → 0.1 V	PA0
8	Load current L2	"	J7-3 / J7-4	"	"	"	PA1
9	Load current L3	"	J7-5 / J7-6	"	"	"	PA2

Neutral lands on J5-4 / J6-4. Voltages are measured **line-to-neutral**; 415 V line-to-line appears between adjacent phases, which is why J5/J6/J15 are 8-pole bodies with only the odd poles wired (10.16 mm live-to-live pitch).

Frequency is not a separate channel. Generator and mains Hz are derived from the L1 voltage waveforms in `ac_sense.c`, by interpolated zero crossings.

CT sizing — confirm before you buy. The schematic and firmware now both specify **50:5**, in one place each (J7 text, and `ECU25_CT_PRIMARY_A` in `firmware/core/ecu_main.h`). Fitting a different ratio without changing that constant scales every current reading by the error. A 25 kVA / 415 V set draws 34.8 A at full load, which is 0.87 A on the secondary — about 17 % of the channel's designed swing, so most of the ADC range goes unused and light-load current is coarse. **50:5** puts full load at ~69 % of span and still does not clip until ~117 A primary (3.4 × overload). Nothing on the board changes; only the part you order. The table above assumes 50:5.

2. Engine sensors — the analog (legacy) path

Three constant-current sender inputs. A current source, not a divider, so the reading does not shift with supply voltage or wiring resistance. Sender types below are the common ones — **calibrate against whatever is actually fitted to your engine**.

#	Quantity	Sensor	Terminal	Range	Excitation	Full scale	MCU pin
10	Oil pressure	resistive pressure sender (VDO / Datcon type, typically 10-184 Ω)	J3-1	0-10 bar	8.06 mA (0.5 V / 62 Ω)	184 Ω → 1.48 V, ≈0.1 Ω resolution	PA3
11	Fuel level	resistive float sender in the tank	J3-2	0-100 %	8.06 mA	sender-dependent	PA4
12	Coolant temperature	NTC thermistor sender	J3-3	-40 to +150 °C	2.0 mA (0.5 V / 249 Ω)	lower current avoids saturating a cold NTC	PA5

J3-4 is the sender return. Each channel has an RC filter plus a 4.7 k series resistor and a BAV199 clamp, holding fault current into the MCU pin to ≈3.6 mA if a sender shorts to +12 V.

#	Quantity	Sensor	Terminal	Range	Front end	MCU pin
13	Engine speed	magnetic pickup on the flywheel ring gear	J4-1 / J4-2	0–4000 rpm	AC-coupled → LM2903 comparator, ±80 mV hysteresis, BAV99 clamps ±50 V pickup swings	PB6 (TIM4_CH1)

3. DC electrical measurement

#	Quantity	Sensor / source	Terminal	Range	Front end	Scaling	MCU pin
14	Battery voltage	direct, at the board's own supply terminal	J1-1	0–18.3 V	100k / 22k divider, BAV199 clamp	÷5.545, 4.5 mV per count	PA6
15	Charge alternator D+	alternator D+ / L terminal	J13	0–18.3 V	120 Ω 3 W excitation + 100k / 22k sense	÷5.545	PA7

D+ is both an input and an output. The 120 Ω resistor feeds the ~100 mA the alternator's field needs to start self-exciting; the same node is then measured to detect charge failure (D+ below `charge_fail_ratio` × battery volts for `charge_fail_delay_ms`).

4. Digital inputs

Eight identical channels on J2 (plus two GND poles). Divider 5.6k / 1.8k from 12 V, BAT54S clamp to the rails, 1 μF debounce cap ($\tau \approx 1.4$ ms), plus a firmware 3-sample debounce. Wetting current ≈1.6 mA — enough to break oxide on a dirty panel contact. Each channel has a solder jumper (JP4–JP11) that fits a 1 k pull-up instead, so the channel accepts a **ground-switched** contact. Fit the pull-up *or* the field 12 V feed, never both.

#	Ch	MCU pin	Firmware field	Sensor / switch	Class
16	DIN1	PE0	emergency_stop	E-stop mushroom button, fail-safe (a broken wire reads as pressed)	shutdown
17	DIN2	PE1	remote_start	External start command — BMS, building controller, dry contact	command
18	DIN3	PE2	low_oil_switch	Engine's own low-oil-pressure switch	shutdown
19	DIN4	PE3	high_coolant_switch	Engine's own high-coolant-temperature switch	shutdown
20	DIN5	PE4	low_coolant_level	Coolant level float switch	warning
21–23	DIN6–8	PE5–PE7	—	Spare — wired to the MCU, no function assigned	—

The mapping is now fixed, in exactly one place: the `DIN_*` defines at the top of `firmware/core/ecu_main.c`. Change it there and change this table; nothing else encodes it. Label the panel to match.

5. Engine data over J1939

Listener only. Every value is aged; once its age passes its timeout the value is marked **invalid** rather than going stale, and the controller falls back or alarms.

PGN	Name	SPN	Quantity	Resolution	Broadcast rate	Timeout
61444	EEC1	190	Engine speed	0.125 rpm/bit	10–20 ms	500 ms
65263	EFL/P1	100	Oil pressure	4 kPa/bit	500 ms	2 s

PGN	Name	SPN	Quantity	Resolution	Broadcast rate	Timeout
65262	ET1	110	Coolant temperature	1 °C/bit, −40 offset	1 s	3 s
65276	DD	96	Fuel level	0.4 %/bit	1 s	5 s
65226	DM1	—	Red (stop) lamp, amber (warning) lamp, first DTC as SPN + FMI	—	1 s	3 s
60928	AC	—	Address claim — contention resolved by lower NAME	—	—	—
59904	RQST	—	Request; answered for the address-claim PGN	—	—	—

Single-frame DM1 only: TP.BAM multi-DTC transfers are not reassembled, but the lamp bits still carry the severity, which is what drives the shutdown. **Comms loss** raises only after run-enable has been on past a 2 s boot grace — with K1 open the engine ECU is asleep and silence is normal.

6. Operator inputs

From the display board over the shared SPI bus (74HC165 key shift register): `key_start`, `key_stop` (which also resets latched alarms), and `mode_auto`.

7. Derived values and alarms

Value	Computed from	Module
RMS volts and amps, all 9 channels	$\sqrt{(\text{mean}(x^2) - \text{mean}(x)^2)}$ over a 100 ms window; DC bias measured, never assumed	<code>ac_sense.c</code>
Generator / mains frequency	interpolated rising zero crossings on the L1 waveform	<code>ac_sense.c</code>
Total real power (signed)	$\text{mean}(v \cdot i) - \text{mean}(v) \cdot \text{mean}(i)$, per phase, summed	<code>ac_sense.c</code>
Power factor	$ P / \Sigma(V_{\text{rms}} \times I_{\text{rms}})$	<code>ac_sense.c</code>
Run hours	ticks with the engine actually turning (rpm > 100), not ticks with K1 closed	<code>runtime</code>
Engine state, AMF state	state machines	<code>engine_fsm.c</code> , <code>amf_fsm.c</code>
21 alarm conditions	thresholds + qualification delays	<code>protection.c</code>

Alarms (each classified warning or shutdown, most with a configurable threshold and delay): `EMERGENCY_STOP`, `LOW_OIL_PRESSURE`, `HIGH_COOLANT_TEMP`, `OVERSPEED`, `UNDERSPEED`, `GEN_UNDER_VOLT`, `GEN_OVER_VOLT`, `GEN_UNDER_FREQ`, `GEN_OVER_FREQ`, `OVERCURRENT`, `CHARGE_FAIL`, `BATT_LOW`, `BATT_HIGH`, `LOW_COOLANT_LEVEL`, `FAIL_TO_START`, `FAIL_TO_STOP`, `SENSOR_LOSS`, `ECU_RED_LAMP`, `ECU_WARNING`, `ECU_COMMS_LOST`.

8. What the controller drives

8.1 Relay outputs

Six G5LE-1-DC12 relays, low-side driven by 2N7002K FETs with SS34 flyback diodes. Coils run from +12V_SW (33.3 mA each, 200 mA total). **Every contact is dry and pilot-duty — no field current crosses the PCB.**

Relay	MCU pin	Net	Contact terminal	Drives
K1 FUEL	PD2	RLY_FUEL	J8-2	Engine ECU run enable (CRDi) or the fuel solenoid (legacy engine)
K2 START	PD3	RLY_START	J8-3	Pilot only — the coil of an external starter relay
K3 GEN	PD4	RLY_GEN	J15 volt-free pair	Generator contactor coil
K4 MAINS	PD5	RLY_MAINS	J15 volt-free pair	Mains contactor coil
K5 AUX1	PD6	RLY_AUX1	J8-4	Configurable — default horn
K6 AUX2	PD7	RLY_AUX2	J8-5	Configurable — default preheat

J8 pins 1 and 6 are both OUT_COM — the common contact supply, which the *installer* feeds from their own fused source. K1/K2/K5/K6 contact commons come from that, not from the board rail. **Ceiling: 4 A total, 2 A per output, installer fuse 4 A.**

START must never drive a starter motor solenoid directly. A 12 V solenoid pulls 20–40 A and glow plugs 28–60 A, against 8 A DC contacts. The contacts would weld — and a welded starter contact means an engine that cranks and cannot be commanded to stop.

K3/K4 are volt-free pairs (COM + NO) on their own block J15, because contactor coils run from their own AC source and the panel's hardware interlock must stay load-bearing.

K5/K6 select from `gcu_aux_fn_t`: `AUX_OFF`, `AUX_HORN`, `AUX_PREHEAT`, `AUX_RUNNING`, `AUX_FAULT`.

8.2 Other outputs

Output	MCU pin	Purpose
D+ excitation	— (analog)	~100 mA into the charge alternator field via 120 Ω 3 W
Display backlight PWM	PE11 (TIM1_CH2)	Dimming; also the crank-time load shed
TFT reset	PE8	RA8875 display controller reset
TFT / LED / key chip selects	PB7 / PB8 / PB9	Shared SPI1 bus to the display board
LED shift register	via PB8	Panel indicator LEDs (74HC595)
EEPROM chip select	PB12	M95M02 config + run hours (SPI2)
RS485 direction	PD10	DE for the Modbus half-duplex turnaround
Heartbeat LED	PA8	Board-alive indicator

8.3 Remote control surface — Modbus RTU

RS485 on J10. Four holding registers, deliberately small; 23 input registers publish the live measurement snapshot read-only.

Reg	Function
0	Mode: 0 off, 1 manual, 2 auto, 3 test
1	Remote start request (0/1)
2	Alarm reset — write 1 to pulse, always reads back 0
3	Lamp test (0/1)

9. What we control on the engine ECU

This is the question worth being precise about, because a common-rail engine ECU manages its own injection, timing, rail pressure, EGR and cold-start aid. **ECU-25 does not touch any of it.**

Physical authority — two contacts:

- **Run enable (K1).** The single most important output. Closed, the engine ECU is permitted to run; open, it stops. On this engine that is an ECU enable input; on a legacy mechanical engine the same contact drives the fuel solenoid. K1 energises from the PREHEAT state onward, not just at crank, to give the ECU its boot time before the starter engages.
- **Starter pilot (K2).** Energises an external starter relay. Crank-disconnect is decided here — on rpm crossing `crank_disconnect_rpm` or oil pressure crossing `crank_disconnect_oil_bar`, whichever comes first.

Over CAN — nothing. ECU-25 transmits only its own J1939 address claim. No TSC1 torque/speed command, no shutdown request, no mode change. Cut the CAN cable and the engine keeps running exactly as before; only the data goes away, and `ECU_COMMS_LOST` raises.

What the engine ECU owns, and we deliberately do not: injection quantity and timing, rail pressure, idle governing, speed droop, EGR, DOC management, glow-plug / cold-start aid (which is why K6 preheat is *configurable* rather than hard-wired), and its own protection shutdowns — those reach us as DM1 lamp bits, and a red lamp makes us open K1.

Everything else the controller commands is switchgear, not engine control: the generator contactor, the mains contactor, and the AMF transfer sequence between them (break-before-make, with a dead time, the panel's mechanical interlock still the final authority).

10. Against a reference AMF controller

The design target is DSE 4520 class — the same functional class as the Kirloskar KG-series controller on the reference sheet. This is a functional checklist of what a controller in that class monitors and switches, and where ECU-25 stands.

The KG640C sheet itself is not in this document's source material, so the comparison is against the standard signal set for the class, not line-by-line against that drawing. Re-share the sheet for a terminal-by-terminal comparison.

Function	Reference class	ECU-25	Note
3-phase generator voltage	yes	yes	3 channels, L-N
3-phase mains voltage	yes	yes	3 channels — AMF needs both sides
3-phase load current	yes	yes	CT inputs, 3 channels
Generator / mains frequency	yes	yes	derived from L1
kW, kVA, power factor	yes	yes	<code>ac_sense.c</code>
Engine speed	yes	yes	magnetic pickup and J1939
Oil pressure	yes	yes	sender and J1939
Coolant temperature	yes	yes	sender and J1939
Fuel level	yes	yes	sender and J1939
Battery voltage	yes	yes	11.0 / 15.5 V thresholds

Function	Reference class	ECU-25	Note
Charge alternator fail	yes	yes	D+ excite + sense
Configurable digital inputs	yes	yes	8, of which 3 spare
Engine ECU J1939 link	yes (CPCB IV+ sets)	yes	listener + DM1 lamps and DTC
Fuel / start / gen / mains relays	yes	yes	all dry contacts
Configurable aux relays	yes	yes	2 (K5 / K6)
Modbus RTU over RS485	yes	yes	FC 03 / 04 / 06 / 16
Run hours	yes	partial	counted, but not yet persisted — the M95M02 driver is unwritten, so hours reset on every power cycle
Event / alarm log	yes	not yet	alarms are live-only; no stored history
Earth / neutral fault CT	some	no	would need a 4th CT channel
Fuel flow / consumption	some	no	no flow sensor input
Battery charger control	some	no	monitoring only
Remote annunciator panel	some	no	Modbus covers remote status
Load sharing / synchronising	no (4520 class)	no	single-set controller by design
DEF / SCR monitoring	N/A on this engine	no	EGR + DOC — a DOC is passive
DPF regeneration UI	N/A on this engine	no	same reason

11. Connector summary

Ref	Poles	Carries
J1	2	Battery + / –
J2	10	DIN1-8 + 2 × GND
J3	4	Oil, fuel, temp senders + return
J4	2	Magnetic pickup
J5	8 (odd wired)	Generator L1 L2 L3 N — 415 V
J6	8 (odd wired)	Mains L1 L2 L3 N — 415 V
J7	6	CT1-3, S1 / S2 each
J8	6	OUT_COM, FUEL, START, AUX1, AUX2, OUT_COM
J9	3	CAN H / L / shield
J10	3	RS485 A / B / shield
J13	2	Charge alternator D+
J15	8 (odd wired)	GEN and MAINS contactor volt-free pairs
J12	20-way ribbon	Display board — SPI, selects, backlight power
J14	3	Debug UART (USART1)

Ref	Poles	Carries
J11	TC2030	SWD programming / debug